

Output_Example:

1. Bipartite Graphs: 5 sets of Bipartite graphs were generated with 5 different scenarios.

Sets of the bipartite graphs:							
	n	m	maxProbability	minCapacity	maxCapacity	maxFlow	scenarios
bg1	10	10	0.2	1	10	26	Small Graph with Low Density
bg2	50	50	0.5	1	100	2286	Moderate Graph with Medium Density
bg3	100	100	1.0	10	500	25202	Large Graph with High Density
bg4	10	1000	0.3	5	20	127	Unbalanced Graph
bg5	1000	1000	0.01	1	50	8376	Sparse Large Graph

Figure 1 The screenshot of Bipartite graphs information of InputGraph/graphGenerationCode/Bipartite/read me.txt

- Ensure that your filePath on line 41 in RunMultipleInputs.java is inputFiles1 (for Bipartite graphs list)

```
RunMultipleInputs.java x
40
41     for (String filePath : inputFiles1) {
42         System.out.println("\nRunning for file: " + filePath);
```

- Recompile the code then run the program to get the outputs

```
akannika@Kannikas-MacBook-Pro src % javac *.java
akannika@Kannikas-MacBook-Pro src % java -cp . RunMultipleInputs

Running for file: InputGraph/graphGenerationCode/Bipartite/bg1.txt
Ford-Fulkerson Runtime: 215 ms
Ford-Fulkerson Maximum Flow: 26
Scaling Ford-Fulkerson Runtime: 5 ms
Scaling Ford-Fulkerson Maximum Flow: 26
Pre-Flow Push Runtime: 2 ms
Pre-Flow Push Maximum Flow: 26

Running for file: InputGraph/graphGenerationCode/Bipartite/bg2.txt
Ford-Fulkerson Runtime: 371 ms
Ford-Fulkerson Maximum Flow: 2286
Scaling Ford-Fulkerson Runtime: 153 ms
Scaling Ford-Fulkerson Maximum Flow: 2286
Pre-Flow Push Runtime: 15 ms
Pre-Flow Push Maximum Flow: 2286

Running for file: InputGraph/graphGenerationCode/Bipartite/bg3.txt
Ford-Fulkerson Runtime: 888 ms
Ford-Fulkerson Maximum Flow: 25202
Scaling Ford-Fulkerson Runtime: 1555 ms
Scaling Ford-Fulkerson Maximum Flow: 25202
Pre-Flow Push Runtime: 114 ms
Pre-Flow Push Maximum Flow: 25202

Running for file: InputGraph/graphGenerationCode/Bipartite/bg4.txt
Ford-Fulkerson Runtime: 572 ms
Ford-Fulkerson Maximum Flow: 127
Scaling Ford-Fulkerson Runtime: 315 ms
Scaling Ford-Fulkerson Maximum Flow: 127
Pre-Flow Push Runtime: 21 ms
Pre-Flow Push Maximum Flow: 127

Running for file: InputGraph/graphGenerationCode/Bipartite/bg5.txt
Ford-Fulkerson Runtime: 241183 ms
Ford-Fulkerson Maximum Flow: 8376
Scaling Ford-Fulkerson Runtime: 2533920 ms
Scaling Ford-Fulkerson Maximum Flow: 8376
Pre-Flow Push Runtime: 168772 ms
Pre-Flow Push Maximum Flow: 8376
akannika@Kannikas-MacBook-Pro src %
```

Figure 2 The screenshot of the output (runtime and maximum flow) of the Bipartite Graphs.

- Fixed Degree Graphs: 5 sets of Fixed Degree graphs were generated with 5 different scenarios.

Sets of FixedDegree Graphs:						
	v	e	minCapacity	maxCapacity	maxFlow	scenarios
fd1	20	10	1	10	61	Small, Dense Graph
fd2	50	15	5	50	409	Moderate-Size Graph with Moderate Connectivity
fd3	100	50	10	100	2552	Large Graph with High Connectivity
fd4	100	5	1	50	134	Sparse Graph with Low Connectivity
fd5	500	10	5	100	441	Very Large Sparse Graph

Figure 3 The screenshot of Fixed Degree graphs information of InputGraph/graphGenerationCode/FixedDegree/readme.txt

- Ensure that your filePath on line 41 in RunMultipleInputs.java is inputFiles2 (for Fixed Degree graphs list)

```

RunMultipleInputs.java x
40
41     for (String filePath : inputFiles2) {
42         System.out.println("\nRunning for file: " + filePath);

```

- Recompile the code then run the program to get the outputs

```

akannika@Kannikas-MacBook-Pro src % javac *.java
akannika@Kannikas-MacBook-Pro src % java -cp . RunMultipleInputs

Running for file: InputGraph/graphGenerationCode/FixedDegree/fd1.txt
Ford-Fulkerson Runtime: 248 ms
Ford-Fulkerson Maximum Flow: 61
Scaling Ford-Fulkerson Runtime: 5 ms
Scaling Ford-Fulkerson Maximum Flow: 61
Pre-Flow Push Runtime: 0 ms
Pre-Flow Push Maximum Flow: 61

Running for file: InputGraph/graphGenerationCode/FixedDegree/fd2.txt
Ford-Fulkerson Runtime: 307 ms
Ford-Fulkerson Maximum Flow: 409
Scaling Ford-Fulkerson Runtime: 52 ms
Scaling Ford-Fulkerson Maximum Flow: 409
Pre-Flow Push Runtime: 303 ms
Pre-Flow Push Maximum Flow: 409

Running for file: InputGraph/graphGenerationCode/FixedDegree/fd3.txt
Ford-Fulkerson Runtime: 536 ms
Ford-Fulkerson Maximum Flow: 2552
Scaling Ford-Fulkerson Runtime: 196 ms
Scaling Ford-Fulkerson Maximum Flow: 2552
Pre-Flow Push Runtime: 29 ms
Pre-Flow Push Maximum Flow: 2552

Running for file: InputGraph/graphGenerationCode/FixedDegree/fd4.txt
Ford-Fulkerson Runtime: 292 ms
Ford-Fulkerson Maximum Flow: 134
Scaling Ford-Fulkerson Runtime: 32 ms
Scaling Ford-Fulkerson Maximum Flow: 134
Pre-Flow Push Runtime: 660 ms
Pre-Flow Push Maximum Flow: 134

Running for file: InputGraph/graphGenerationCode/FixedDegree/fd5.txt
Ford-Fulkerson Runtime: 678 ms
Ford-Fulkerson Maximum Flow: 441
Scaling Ford-Fulkerson Runtime: 364 ms
Scaling Ford-Fulkerson Maximum Flow: 441
Pre-Flow Push Runtime: 27 ms
Pre-Flow Push Maximum Flow: 441
akannika@Kannikas-MacBook-Pro src %

```

Figure 4 The screenshot of the output (runtime and maximum flow) of the Fixed Degree Graphs.

3. Mesh Graphs: 5 sets of Mesh graphs were generated with 5 different scenarios.

Sets of Mesh Graphs:					
	m	n	c	maxFlow	scenarios
mg1	3	4	1	3	Small, Fixed Capacity
mg2	4	5	10	17	Moderate, Random Capacity
mg3	6	6	5	16	Larger, Fixed Capacity
mg4	8	8	20	70	Larger, Random Capacity
mg5	10	10	100	273	Very Large, Fixed Capacity

Figure 5 The screenshot of Mesh graphs information of InputGraph/graphGenerationCode/Mesh/readme.txt

- Ensure that your filePath on line 41 in RunMultipleInputs.java is inputFiles3 (for Mesh graphs list)

```

RunMultipleInputs.java x
40
41     for (String filePath : inputFiles3) {
42         System.out.println("\nRunning for file: " + filePath);

```

- Recompile the code then run the program to get the outputs

```

akannika@Kannikas-MacBook-Pro src % javac *.java
akannika@Kannikas-MacBook-Pro src % java -cp . RunMultipleInputs

Running for file: InputGraph/graphGenerationCode/Mesh/mg1.txt
Ford-Fulkerson Runtime: 209 ms
Ford-Fulkerson Maximum Flow: 3
Scaling Ford-Fulkerson Runtime: 2 ms
Scaling Ford-Fulkerson Maximum Flow: 3
Pre-Flow Push Runtime: 0 ms
Pre-Flow Push Maximum Flow: 3

Running for file: InputGraph/graphGenerationCode/Mesh/mg2.txt
Ford-Fulkerson Runtime: 219 ms
Ford-Fulkerson Maximum Flow: 17
Scaling Ford-Fulkerson Runtime: 4 ms
Scaling Ford-Fulkerson Maximum Flow: 17
Pre-Flow Push Runtime: 1 ms
Pre-Flow Push Maximum Flow: 17

Running for file: InputGraph/graphGenerationCode/Mesh/mg3.txt
Ford-Fulkerson Runtime: 233 ms
Ford-Fulkerson Maximum Flow: 16
Scaling Ford-Fulkerson Runtime: 8 ms
Scaling Ford-Fulkerson Maximum Flow: 16
Pre-Flow Push Runtime: 2 ms
Pre-Flow Push Maximum Flow: 16

Running for file: InputGraph/graphGenerationCode/Mesh/mg4.txt
Ford-Fulkerson Runtime: 261 ms
Ford-Fulkerson Maximum Flow: 70
Scaling Ford-Fulkerson Runtime: 40 ms
Scaling Ford-Fulkerson Maximum Flow: 70
Pre-Flow Push Runtime: 157 ms
Pre-Flow Push Maximum Flow: 70

Running for file: InputGraph/graphGenerationCode/Mesh/mg5.txt
Ford-Fulkerson Runtime: 283 ms
Ford-Fulkerson Maximum Flow: 273
Scaling Ford-Fulkerson Runtime: 41 ms
Scaling Ford-Fulkerson Maximum Flow: 273
Pre-Flow Push Runtime: 263 ms
Pre-Flow Push Maximum Flow: 273
akannika@Kannikas-MacBook-Pro src %

```

Figure 6 The screenshot of the output (runtime and maximum flow) of the Mesh Graphs.

4. Random Graphs: 5 sets of Random graphs were generated with 5 different scenarios.

Sets of Random graphs:						
	v	density	min	max	maxFlow	scenarios
rg1	10	low-10%	1	10	4	Low density, 10 vertices, capacity 1-10
rg2	10	med-30%	1	20	6	Medium density, 10 vertices, capacity 1-20
rg3	10	high-50%	1	30	59	High density, 10 vertices, capacity 1-30
rg4	20	low-10%	1	10	7	Low density, 20 vertices, capacity 1-10
rg5	20	high-70%	1	50	371	High density, 20 vertices, capacity 1-50

Figure 7 The screenshot of Random graphs information of InputGraph/graphGenerationCode/Random/Read me.txt

- Ensure that your filePath on line 41 in RunMultipleInputs.java is inputFiles4 (for Random graphs list)

```

RunMultipleInputs.java ×
40
41     for (String filePath : inputFiles4) {
42         System.out.println("\nRunning for file: " + filePath);

```

- Recompile the code then run the program to get the outputs

```

akannika@Kannikas-MacBook-Pro src % javac *.java
akannika@Kannikas-MacBook-Pro src % java -cp . RunMultipleInputs

Running for file: InputGraph/graphGenerationCode/Random/rg1.txt
Ford-Fulkerson Runtime: 202 ms
Ford-Fulkerson Maximum Flow: 4
Scaling Ford-Fulkerson Runtime: 0 ms
Scaling Ford-Fulkerson Maximum Flow: 4
Pre-Flow Push Runtime: 0 ms
Pre-Flow Push Maximum Flow: 4

Running for file: InputGraph/graphGenerationCode/Random/rg2.txt
Ford-Fulkerson Runtime: 201 ms
Ford-Fulkerson Maximum Flow: 6
Scaling Ford-Fulkerson Runtime: 1 ms
Scaling Ford-Fulkerson Maximum Flow: 6
Pre-Flow Push Runtime: 3 ms
Pre-Flow Push Maximum Flow: 6

Running for file: InputGraph/graphGenerationCode/Random/rg3.txt
Ford-Fulkerson Runtime: 208 ms
Ford-Fulkerson Maximum Flow: 59
Scaling Ford-Fulkerson Runtime: 3 ms
Scaling Ford-Fulkerson Maximum Flow: 59
Pre-Flow Push Runtime: 0 ms
Pre-Flow Push Maximum Flow: 59

Running for file: InputGraph/graphGenerationCode/Random/rg4.txt
Ford-Fulkerson Runtime: 207 ms
Ford-Fulkerson Maximum Flow: 7
Scaling Ford-Fulkerson Runtime: 1 ms
Scaling Ford-Fulkerson Maximum Flow: 7
Pre-Flow Push Runtime: 9 ms
Pre-Flow Push Maximum Flow: 7

Running for file: InputGraph/graphGenerationCode/Random/rg5.txt
Ford-Fulkerson Runtime: 252 ms
Ford-Fulkerson Maximum Flow: 371
Scaling Ford-Fulkerson Runtime: 5 ms
Scaling Ford-Fulkerson Maximum Flow: 371
Pre-Flow Push Runtime: 0 ms
Pre-Flow Push Maximum Flow: 371
akannika@Kannikas-MacBook-Pro src %

```

Figure 8 The screenshot of the output (runtime and maximum flow) of the Random Graphs.